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22F-3277
BCS-3C
LAB 7

Q3. Write a Subroutine that calculates the factorial of a given number using following requirements.

- **Number is 10**
- **Taking all parameter from the stack**
- **Saving and restoring register values**
- **All stack must be cleared before ending the program**
- **Mnemonic “mul” is not allow to use.**

```
org 0x0100
```

```
jmp start
```

```
hi:
```

```
    push bp
    mov bp,sp
    sub sp,2

    mov bx, [bp+4]
```

```
factorial:
```

```
    dec bx
    mov dx, bx
    mov cx, 8
```

```
l1:
```

```
    shr dx, 1
    jnc skip
```

```
    add [bp-2], ax
```

```
skip:
```

```
    shl ax, 1
    dec cx
    jnz l1
```

```
    cmp bx, 1
    je done
    mov ax, [bp-2]
    mov word [bp-2], 0
```

```
done:
```

```
cmp bx, 1
jne factorial
```

```
pop bp
mov sp,bp
pop bp
ret
```

start:

```
mov ax , num
push ax
call hi
```

```
mov ax, 0x4c00
int 0x21
```

```
num: db 8h
result: db 0
```

CPU 0000										CPU 0000										CPU 0000										CPU 0000									
BX 0000										DI 0000										DS 19F5										ES 19F5									
CX 0000										BP 0000										ES 19F5										HS 19F5									
DX 0000										SP FFFE										SS 19F5										FS 19F5									
[R3] reg=0000										+2 20CD										+4 9FFD										OF DF IF SF ZF AF PF CF									
CMD >SSS										0008										1										0 1 2 3 4 5 6 7									
0100 A13201										MOV AX, [0132]										DS:0000										CD 20 FF 9F 00 EA FF FF									
0103 89C3										MOV BX, AX										DS:0008										AD DE 1B 05 C5 06 00 00									
0105 4B										DEC BX										DS:0010										18 01 10 01 18 01 92 01									
0106 89DA										MOV DX, BX										DS:0018										01 01 01 00 FF 00 01 FF									
0108 B91000										MOV CX, 0010										DS:0020										FF FF FF FF FF FF FF FF									
010B D1EA										SHR DX, 1										DS:0028										FF FF FF FF FF 19 E4 11									
010D 7304										JNC 0113										DS:0030										A2 01 14 00 EB 00 F5 19									
010F 01063401										ADD [0134], AX										DS:0038										FF FF FF FF 00 00 00 00									
																				DS:0040										05 00 00 00 00 00 00 00									
																				DS:0048										00 00 00 00 00 00 00 00									

Q4. Write assembly code to implement bubble sort with mentioned following requirements.

- Use subroutine for calling
- Use a local variable(stack) to store the swap flag.
- Taking all parameter from the stack
- Saving and restoring register values
- All stack must be cleared before ending the program
- array1: dw 80, -7, -2, 45, 81, 50, 10, 9, 30, 99, 0
- array2: dw 70, 32, 21, -3, -1, 31, 44, -9, 32

```
[org 0x0100]
```

```
jmp start
```

```
array1: dw 80h,-7h,-2h,45h,81h,50h,10h,9h,30h,99h,0h
```

```
array2: dw 70h,32h,21h,-3h,-1h,31h,44h,-9h,32h
```

```
bubblesort:
```

```
    push bp
    mov bp, sp
    sub sp, 2
    push ax
    push bx
    push cx
    push si
```

```
    mov bx, [bp+6]
    mov cx, [bp+4]
    dec cx
    shl cx, 1
```

```
mainloop:
```

```
    mov si, 0
    mov byte [bp-2], 0
```

```
innerloop:
```

```
    mov ax, [bx+si]
    cmp ax, [bx+si+2]
    jle noswap
```

```
    xchg ax, [bx+si+2]
    mov [bx+si], ax
    mov byte [bp-2], 1
```

```
noswap:
```

```
    add si, 2
    cmp si, cx
    jne innerloop
```

```
    cmp byte [bp-2], 1
    je mainloop
```

```
    pop si
    pop cx
```

```

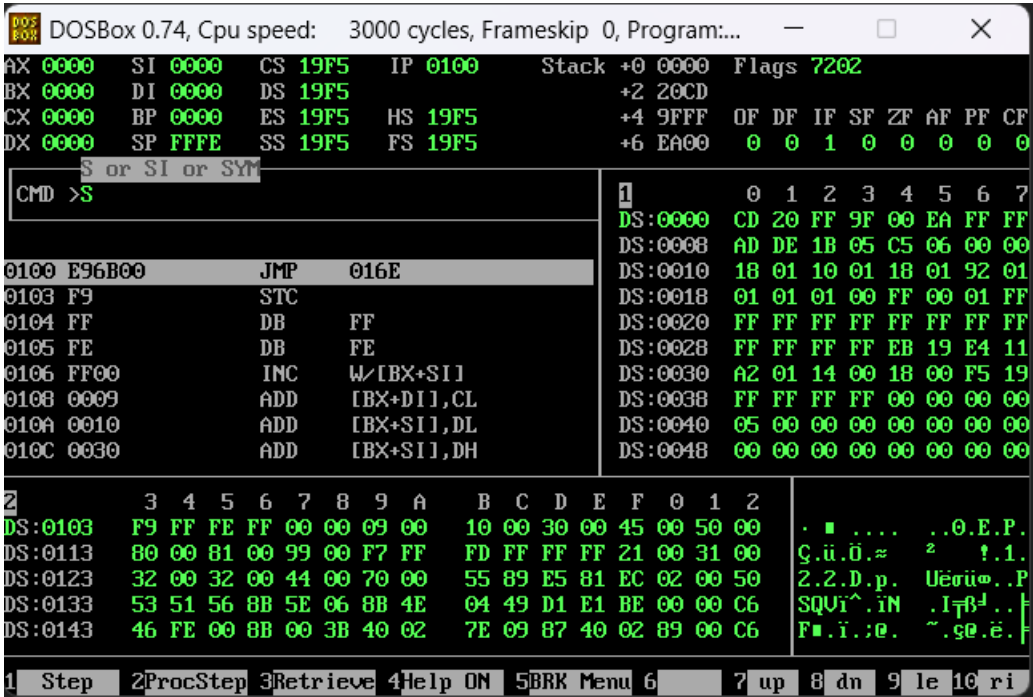
pop bx
pop ax
mov sp, bp
pop bp
ret 4

start:
    mov ax, array1
    push ax
    mov ax, 11
    push ax
    call bubblesort

    mov ax, array2
    push ax
    mov ax, 9
    push ax
    call bubblesort

    mov ax, 0x4c00
    int 0x21

```



Array1 starts from 0103
 Array2 starts from 0119